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Global heterochromatin loss

A unifying theory of aging?

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The aging field is replete with theories. Over the past years, many distinct, yet overlapping mechanisms have been proposed to explain organismal aging. These include free radicals, loss of heterochromatin, genetically programmed senescence, telomere shortening, genomic instability, nutritional intake and growth signaling, to name a few. The objective of this Point-of-View is to highlight recent progress on the "loss of heterochromatin" model of aging and to propose that epigenetic changes contributing to global heterochromatin loss may underlie the various cellular processes associated with aging.

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Epigenetic Changes during Aging and the "Loss of Heterochromatin" Model

The eukaryotic genome is organized in a highly ordered chromatin structure. The nucleosome is the most fundamental unit of chromatin and consists of 146–7 base pairs of DNA wrapped around an octomer of the major core histones H2A, H2B, H3 and H4.[1,2] Histones are subject to a multitude of post-translational covalent modifications such as methylation, acetylation, phosphorylation, ubiquitination and sumoylation at numerous residues that contribute to changes in chromatin conformation and may serve as recognition sites for chromatin regulators with chromo, tudor, WD40 repeat or plant homeodomain (PHD) finger domains.[3,4] It has been established that different regions of the eukaryotic genome exist in either the relatively loose euchromatin or tightly packaged heterochromatin state. Euchromatin regions are transcriptionally active domains in contrast to the transcriptionally silent heterochromatin regions, although there are exceptions.[5,6] Histone hypoacetylation, H3K9 hypermethylation and heterochromatin protein 1 (HP1) recruitment are hallmark features of heterochromatin formation.[7,8] Heterochromatin is further divided into constitutive heterochromatin, such as in centromeric and telomeric segments that adopt the silent formation in all cell types and facultative heterochromatin as found in the inactivated X chromosome, present in only a subset of cells.[9-11]

The "loss of heterochromatin" model of aging was first put forth by Villeponteau (1997), who proposed that heterochromatin domains established early in embryogenesis are broken down during the aging process, contributing to the derepression of silenced genes and leading to aberrant gene expression patterns.[12] Consistent with this proposal, global heterochromatin loss has been associated with aging in humans and animals. For example, in humans, germline mutations in lamins cause Hutchinson-Gilford progeria syndrome (HGPS) and atypical Werner syndrome, diseases that mimic premature aging.[13] Cultured cells obtained from a female HGPS patient and expressing progerin, a truncated, mutant form of Lamin A, exhibit abnormal nuclear morphology indicative of a loss of heterochromatin.[13] The cells also show a decrease in the heterochromatin marker H3K9me3 and a delocalization of HP1α and the CREST antigen from

constitutive heterochromatin. Consistent with the loss of pericentric heterochromatin, pericentric satellite III repeat transcription is derepressed. With regard to facultative heterochromatin, downregulation of the EZH2 methyltransferse and concomitant loss of H3K27me3 occur at the inactivated X-chromosome. Although an increase in the constitutive heterochromatic marker H4K20me3 was also observed, these results overall point to the disruption of heterochromatin in aging-related phenotypes.

These findings were substantiated in studies of natural aging in *C. elegans*, Drosophila and humans. A similar change in nuclear architecture and a loss of peripheral heterochromatin were detected in nonneuronal cells during aging of *C. elegans*.[14] In Drosophila, a gradual loss of heterochromatin correlates with age and lamin proteins were found to have a role in aberrant nuclear morphology and reduced lifespan.[15,16] In humans, skin fibroblast cell lines established from old individuals showed nuclear defects similar to those of an HGPS patient's cells, including changes in histone modifications and increased DNA damage.[17] Interestingly, in cells cultured from healthy individuals, inhibiting the sporadic use of a cryptic splice site in lamin A that produces progerin reverses the nuclear defects associated with aging, further demonstrating that the global loss of heterochromatin markers and dissociation of heterochromatin proteins seen in HGPS also occur during physiological aging.[17]

One may argue that the "loss of heterochromatin" model of aging is perhaps an over-simplification, since simultaneously with the global decrease in constitutive heterochromatin observed with aging, an increase in localized facultative heterochromatin occurs in the form of senescence-associated heterochromatin foci (SAHFs). SAHF formation is a hallmark of cellular senescence, which is associated with molecular mechanisms that contribute to irreversible cell cycle arrest, such as telomere shortening, downregulation of the canonical Wnt signaling pathway, upregulation of the cyclin dependent kinase inhibitor $p16^{INK4a}$, activation of ataxia-telangiectasia mutated (ATM) kinase and p53, to name just a few key events found in late passage cultured cells and in aged animal cells, including those of primates.[18-22] Studies have shown that SAHF formation occurs in a sequential order, in which chromosomes first decondense, followed by H3K9 methylation, binding of HP1 proteins and finally, inclusion of macroH2A; this process requires key players in heterochromatin formation including histone chaperones HIRA and Asf1, HMGA proteins, H3K9 methyl-transferases Suv39 h1 and h2, and heterochromatin proteins HP1α, β and γ.[21-26] Sites of SAHF formation include E2F target genes at which recruitment of heterochromatin-inducing and silencing proteins depends on the Rb pathway.[27] Moreover, the establishment of oncogene-induced SAHF requires Ataxia-telangiectasia and Rad3-related (ATR) and the transcriptional repression of E2F target genes requires ATM and p53.[28,29] Thus, while there is a decrease in "global" heterochromatin levels, aging is also associated with an increase in localized heterochromatin formation at specific loci. This process has been referred to as "heterochromatin redistribution."[26]

Further support for the global loss of heterochromatin model of aging comes from the observation that DNA methylation, which is associated with heterochromatin formation, also becomes depleted globally with aging, particularly at constitutively heterochromatic, repetitive DNA sequences.[30-33] Similarly to heterochromatin redistribution, DNA methylation in senescent cells becomes enriched at gene-specific promoter sequences, particularly those that play important roles in senescence and cancer, such as ERα, IGF-II, $P14^{ARF}$, $p16^{INK4A}$, Ecadherin, c-fos and collagen-α.[34-42] The redistribution of global DNA methylation has been attributed to the differential regulation of the Dnmt1 DNA methyltransferase that maintains DNA methylation patterns by methylating hemimethylated sites, in contrast to Dnmt3a and Dnmt3b, which are responsible for de novo DNA methylation.[43] In aging and immortalized fibroblasts, Dnmt1 expression and activity decrease, concomitantly with an upregulation of Dnmt3b transcription.[44,45] DNA methylation serves as a docking site for Methyl-CpG-binding proteins (MBPs) that recruit transcriptional co-repressors for gene silencing and chromatin remodeling. Furthermore, DNMTs interact with H3K9 methyltransferases, HP1 and histone deacetylases.[46-57] At the rDNA locus, the N-CoR repressor complex, DNA methyltransferases and histone deacetylases assemble via protein-protein interaction in a sequential manner to mediate transcriptional repression of rRNA, illustrating the intimate link between DNA methylation, H3K9 methylation and histone deacetylation culminating in heterochromatin formation.[58,59]

Changes in histone modifications have also been associated with aging. Utilizing the brains of senescence-accelerated prone mouse 8 (SAMP8) as a model for aging, decreases were found in H4K20me and H3K36me3 in the brains of aged mice, while there were increases in H3K27me3, H3K79me and H3K79me2.[60] In the kidneys and livers of aged rats, however, H4K20me3, a marker for repression of gene transcription, increased with aging, while H4K20me1,2 levels did not change.[61] H3K27me3, a marker that serves as a recognition site for the polycomb repressive complex (PRC), transcriptionally silences the INK4/ARF locus in young cells, however in aging cells the H3K27 demethylase JMJD3 is upregulated and the locus is derepressed, consistent with a senescent phenotype.[62] These observations are in accord with those seen in SAHF of senescent cells, as described earlier, where histone methyltransferases, demethylases and the retinoblastoma family of proteins play integral roles in epigenetic regulation of cell cycle gene promoters. Thus, changes in epigenetic signatures during the aging process require the differential expression of specific chromatin modifiers.[63-65]

In summary, it appears that aging is associated with an overall net decrease in heterochromatin, but accompanied by an increase in heterochromatin at specific loci. The "loss of global heterochromatin" model for aging may, therefore, best be described as a net decrease in constitutive heterochromatin, with a partial redistribution of heterochromatin markers to facultative heterochromatin, resulting in the altered gene expression patterns associated with aging (**Fig. 1**).

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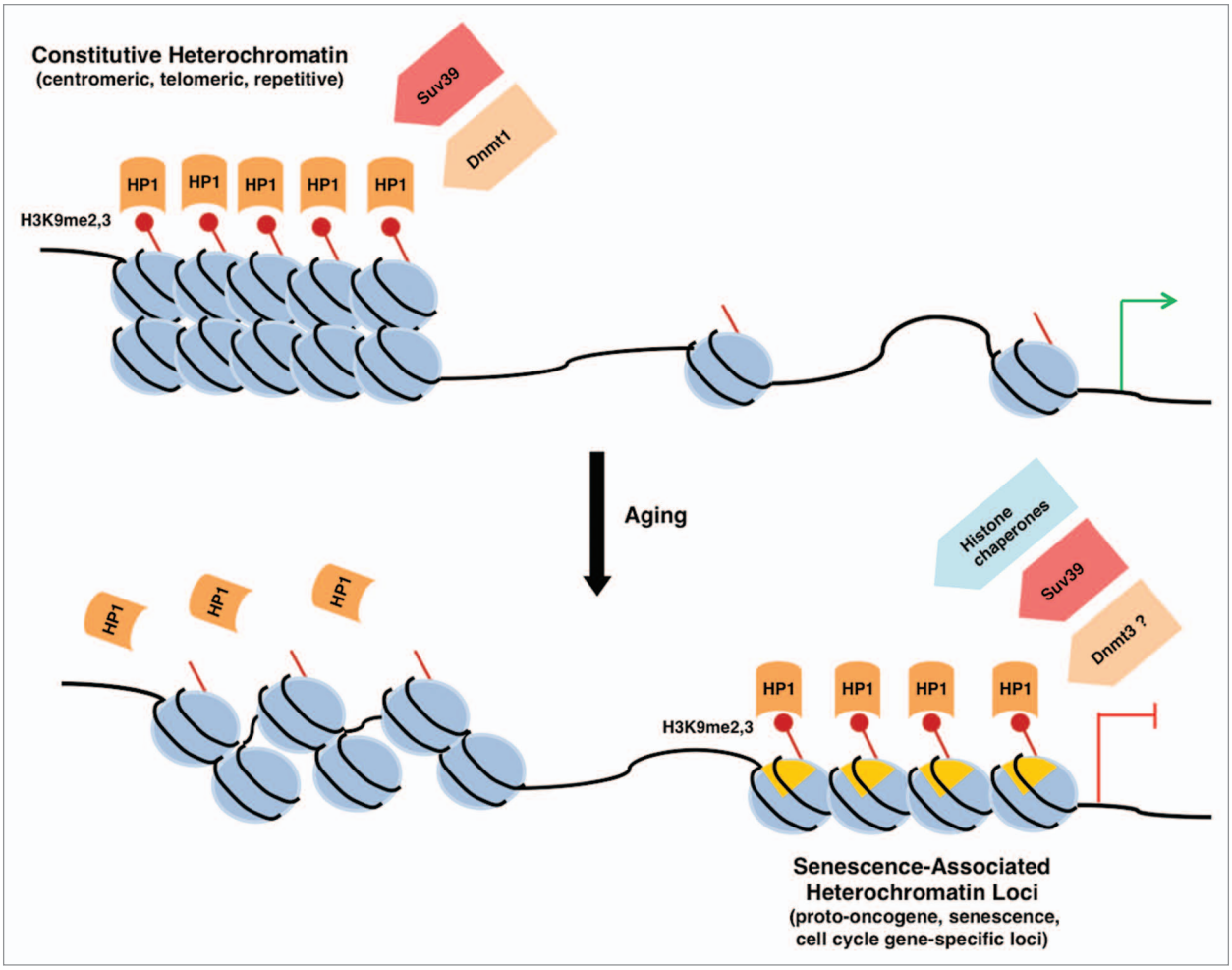


Figure 1. Heterochromatin redistribution during aging. Aging is associated with a loss of constitutive heterochromatin, as demonstrated by a decrease in H3K9 methylation and delocalization of HP1 proteins. Concurrently, increases in facultative heterochromatin occur at specific loci, particularly at senescence-associated heterochromatin foci (SAHFs) that depend on the activity of H3K9 methyltransferases, on recruitment of HP1 proteins and histone chaperones, and on the presence of macro H2A. The decrease in activity of the maintenance DNA methyltransferase Dnmt1 and the increase in de novo Dnmt3 methyltransferases during aging may also contribute to heterochromatin redistribution.

Linking the Various Models of Aging and Associated Molecular Mechanisms: The Central Role of Heterochromatin in Aging

Recently, it has been found that overexpressing heterochromatin protein 1a (HP1a) in Drosophila prolongs lifespan and slows the breakdown of muscle integrity that is characteristic of old animals.[16,66] On the other hand, loss-of-function HP1a mutant animals have a shortened lifespan. A study of epigenetic signature changes during normal aging in Drosophila showed an overall decrease in H3 and the heterochromatin marker H3K9me2 as well as HP1a delocalization, similar to the findings in other organisms.[13,14,17,67] These results support the global heterochromatin loss model of aging, since modulating HP1a levels affect aging and its associated phenotypes. The complex molecular interactions associated with aging have given rise to various aging models including ROS insult, genomic instability, genetically programmed senescence, telomere shortening, nutritional intake and growth signaling. All of these models could have heterochromatin alteration as an underlying theme.

Further molecular characterization demonstrated that with the loss-of-function HP1a genotype, or a JAK gain-of-function genotype, each of which perturbs heterochromatin,[68,69] nucleolar morphology was disrupted and formation of extrachromosomal circular (ECC) DNA increased, reflecting recombination due to rDNA instability.[16] Conversely, overexpressing HP1a or a non-phosphorylatable STAT, which increases heterochromatin stability,[69,70] extended life span.[16] These results provide evidence for aging mechanisms mediated by heterochromatin modulation, as discussed below.

Yeast replicative aging and Sir2. While the importance of heterochromatin in nucleolar stability has been demonstrated in flies,[71,72] the role of sirtuin histone deacetylases in rDNA stability has been studied extensively in *S. cerevisiae* in the context of normal aging. Sir2 deletion shortens life span in yeast and is associated with increased rDNA recombination, while providing an extra genomic copy of Sir2 extends lifespan and enhances rDNA stability.[73-76] Moreover,

it has been observed that during aging, the Sir2-containing protein complex is redistributed from telomeres to the rDNA locus, where silencing occurs by means of heterochromatin formation mediated by Sir2 and the Sir2-associated proteins, Sir3 and Sir4.[74,77,78] In line with the idea of "heterochromatin redistribution" mentioned earlier, Oberdoerffer and Sinclair (2007) propose that the redistribution of heterochromatin-associated factors is a major mechanism leading to the changes in nuclear architecture and gene expression patterns associated with aging.[78]

Sirtuins in other organisms. Anti-aging effects of Sir2 orthologs in various other organisms including Drosophila, *C. elegans*, and mammals have been reported, although direct involvement in rDNA silencing during the aging process has yet to be definitively determined.[79,80] Pharmacological inhibition of Sir2 decreases lifespan in various organisms and the expression of SIRT1, the mammalian Sir2 ortholog, decreases with age in mitotic tissues of normal aging mice and in murine and human cell culture.[81,82] SIRT1 levels have been found to correlate with proliferating cell nuclear antigen (PCNA) levels, while being inversely correlated with senescence-activated β-galactosidase activity and p63-mediated senescence.[81,82] These findings reinforce the idea that this epigenetic silencer has a role in organismal aging.[83-85] However, one recent study has challenged the validity of some previous studies showing anti-aging effects of Sir2 homologs in *C. elegans* and Drosophila.[86] Thus, further work is required to determine whether the role of Sir2 orthologs in promoting longevity is generally conserved.

Caloric restriction. The heterochromatin-inducing histone deacetylase, Sir2, has been extensively studied with regard to its involvement in caloric restriction (CR), a calorie-reducing dietary regimen that has been shown to extend lifespan in yeast, worms, flies and mammals.[87] One of the major mechanisms by which CR extends lifespan appears to involve Sir2/SIRT1 activation, although other mechanisms, including decreased Tor signaling[88] and the generation of reactive oxygen species (ROS)[89] have been implicated as well.[84,85,90-93] When nutrients are scarce, yeast changes metabolic activity leading to altered NAD/NADH ratio, thus increasing the availability of NAD, the required catalytic cofactor of Sir2.[91,94-96] Under CR conditions in yeast, Sir2 or its homolog, HST2 suppresses rDNA recombination although this idea has sparked controversy and is still under debate.[97-100] It has been reported that lifespan in flies can be prolonged by the Sir2 activator resveratrol or by overexpression of Sir2, and under these conditions, lifespan is not further extended by CR tand therefore it is speculated that Sir2 activation may be the main downstream effector of CR.[84,85] The direct involvement of Sir2 in maintaining rDNA stability and preventing ECC formation, however hasnot been demonstrated in flies. In mammals, SIRT1 and heterochromatin silence rDNA transcription in response to low intracellular energy levels.[101] Taken together with the observations in flies that loss-of-function of HP1a produces larger flies, while overepression of HP1a leads to smaller flies,[16] these studies suggest that enhanced rRNA transcription resulting from a decrease in heterochromatin at the rDNA locus may increase protein synthesis and growth to accelerate aging.

CR-induced SIRT1 and HDAC1 activation can also contribute to longevity by transcriptionally downregulating genes involved in senescence including $p16^{INK4a}$, hTERT, SFRP1 and E-Cadherin, by deacetylating H4K16 at their genomic loci.[102,103] Moreover, SIRT1 mediates functional cross-talk between histone deacetylation and the heterochromatin marker H3K9me3. While deacetylating H4K16, SIRT1 simultaneously upregulates activity of the SUV39H1 methyltransferase by deacetylating a residue in its catalytic SET domain.[104] CR also leads to altered DNA methylation patterns. SIRT1 deacetylates the DNA methyl transferase DNMT1 to regulate its function, indicating the existence of an intricate network of epigenetic cross-talk.[105] Studies in rats, mice and a cultured human cell model of CR provide examples of regulation by DNA methylation. In the cultured human cell CR model, DNA hypermethylation occurs at the E2F-1 binding sequence within the $p16^{INK4a}$ promoter and leads to its downregulation.[106] In pancreatic acinar cells of rats, CR results in increased promoter DNA methylation of the Ras proto-oncogene.[107] In mice, CR is able to limit the increase in Dnmt3a levels that normally occurs in aging cells.[108,109] Thus, nutrition and growth signaling both affect aging, mediated at least in part by heterochromatin redistribution. Regulation, of the expression of genes involved in metabolism, apoptosis and cellular senescence by increased local facultative heterochromatinization may be an important mechanism by which aging is regulated.

In contrast to yeast Sir2, which appears to deacetylate only histones, mammalian SIRT1 can directly deactylate the lysines of multiple target proteins including PGC-1α, p53 and FOXO, to confer oxidative stress resistance, and the DNA repair protein, Ku70, leading to inactivation of Bax and inhibition of apoptosis.[73,110-112] These results suggest that apart from epigenetic modulation, SIRT1 can promote longevity in multiple ways, linking nutritional effects, genomic stability and heterochromatin protection.

Insulin signaling. In *C. elegans*, genetic analysis has demonstrated that sir-2 acts within the insulin-like receptor (IGF) pathway.[83] Insulin signaling and factors promoting growth and protein synthesis tend to accelerate aging, while inhibition of this pathway extends life span. Mutations in the Insulin/IGF-1-like receptor pathway have been found to extend lifespan in *C. elegans*, Drosophila and mice. Studies show that increased longevity resulting from decreased Insulin/IGF signaling is mediated to a great extent by FOXO and on the other hand, insulin/IGF pathway accelerates aging by signaling to TOR.[113-118] The insulin-like receptor pathway has a large number of downstream transcriptional targets through which it may affect aging. These targets, which have been identified in various organisms, include antioxidant, chaperone, apolipoprotein, amino acid turnover and antibacterial genes, such as superoxide dismutase, metallothionine, catalase and glutathionine S-transferase,[117,119-124] and collectively suggest that nutritional intake and growth signaling are related to ROS and DNA damage.

The insulin/IGF-1 signaling pathway is also directly linked to heterochromatin

regulation during organismal aging in *C. elegans*, where the abnormal nuclear morphology phenotype associated with heterochromatin disruption and aging are delayed by insulin/IGF-1 signaling mutations.[14] Furthermore, in normally aging human cells, inhibiting the sporadic use of a splice site producing progerin, the truncated form of lamin A causing HGPS, leads to downregulation of *p21*, *IGFBP3* and *GADD45B*.[17] Taken together, these observations demonstrate that lifespan extension by reduced growth signals is tightly coupled to the maintenance of heterochromatin domains.

Free radicals. Perhaps one of the most prominent theories of aging is the "free radical theory," proposed by Harman in the 1950s. Free radicals can be intrinsic ROS that are normal byproducts of metabolism, or extrinsic ROS, such as those generated by radiation. ROS are sources of damage to macromolecules (they cause lipid peroxidation, protein damage, and single- and double-strand breaks, adducts and cross-linking of DNA), and their accumulation contributes to the aging process.[90,125] Consistently with this idea, 8-oxoguanine (oxo8dG), a major product of oxidative damage to DNA, accumulates with age, and many longevity-promoting mutations in model organisms increase resistance to oxidative stress.[126,127] A study using human brain tissue and cultured neurons suggests that an accumulation of oxidative DNA damage at the promoters of genes involved in neural function may account for age-related cognitive decline.[128] However, the relationship between ROS and aging appears to be more complex than the free radical theory suggests. Reactive oxygen and nitrogen species have been found to function in normal cell signaling and in some studies on CR lifespan extension in yeast, flies and worms, increased oxidative stress was detected in longer-lived animals.[90] Additionally, mice heterozygous for Sod2 have lifespans comparable to wildtype despite increased levels of nuclear and mitochondrial 8oxodG with age.[129] Indeed, the hormesis theory of aging argues that low-level stimulation by stress signals increases lifespan.[3] Nevertheless, it is evident that a sufficient accumulation of damage to macromolecules accelerates aging and the onset of aging-related pathologies. There is emerging evidence suggesting a protective role of heterochromatin in counteracting ROS-induced DNA damage, as further discussed below.

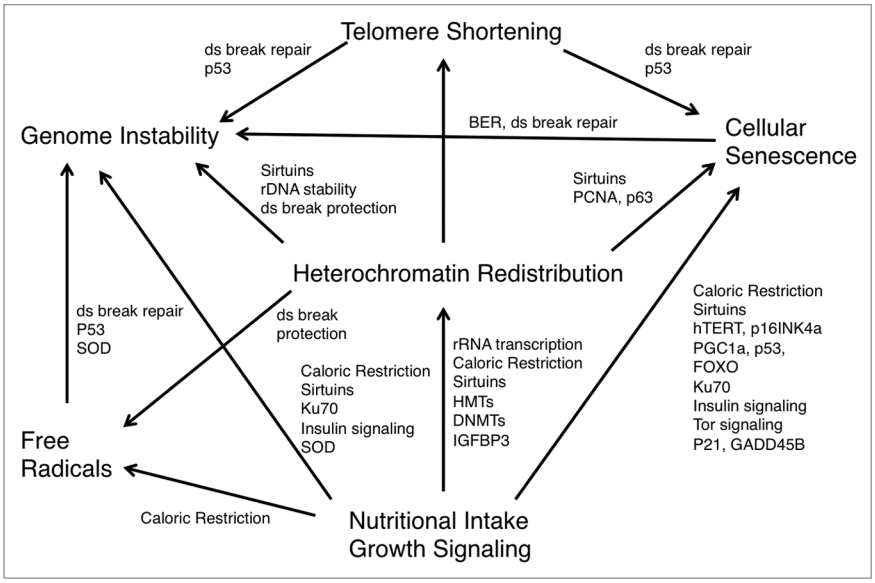


Figure 2. Epigenetic changes resulting in a decrease and redistribution of global heterochromatin may underlie the various models of aging. The models of free radical accumulation, genetically programmed senescence, telomere shortening, genomic instability, nutritional intake and growth signaling are distinct, yet overlapping theories of aging that may all be linked by heterochromatin redistribution.

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Genome instability. Genomic damage that results in genomic instability is a major inducer of the DNA damage response and p53-mediated senescence, as described earlier. Defects in the DNA repair machinery as a consequence of germline mutations in ATM and ATR, for instance, can cause phenotypes resembling premature aging.[130-133] As with damage caused by factors such as ROS, DNA damage caused by replication errors and by spontaneous chemical changes to the DNA can also lead to genomic instability and senescence. In Drosophila, it has been demonstrated that increased levels of HP1a or unphosphorylated STAT, which augment global heterochromatin levels, provide protection against double-stranded DNA breaks induced by ionizing radiation and enhance survival.[134] Conversely, reducing heterochromatin levels increases sensitivity to DNA damage by ionizing radiation and increases the incidence of radiation-induced cell cycle arrest.[134] It is therefore plausible to speculate that sustaining the global level of heterochromatin at relatively high levels may contribute to extended lifespan by protecting against DNA damage.

Telomere shortening. Telomere shortening is another cellular signal that induces senescence, by way of ATM, Chk2 and activation of p53.[135-139] In *S. cerevisiae*, Sirtuins that are associated with telomeric heterochromatin are integral to telomere stability.[140-142] At telomeres of telomerase deficient mice with an accelerated aging phenotype, or in late passage cultured cells,[143,144] the heterochromatin markers H3K9me3, H4K20me3 and CBX3 diminish, while the euchromatin marker H3K9ac increases.[145,146] On the other hand, levels of the telomere binding proteins TERF1 and TERF2 remain stable, demonstrating the specificity of epigenetic changes in this region. DNA methylation at telomeres also declines under these conditions, consistent with previous findings that mouse cells deficient for DNMTs have elongated telomeres that show increased recombination.[147] Thus loss of heterochromatin at telomeric regions and the resulting instability is another factor contributing to the aging process.

Cellular senescence. Senescent cells accumulate upon organismal aging and in the tissues of patients with pathologic conditions that mimic premature aging.

Although senescence can occur in some cells without SAHF formation, in mouse lymphocytes, a lack of H3K9 methylation and Suv39 h1 prevent Ras-induced senescence, suggesting that heterochromatin redistribution and SAHF formation may be upstream regulators of senescence.[65,148,149] Although cellular senescence and organismal aging are tightly associated, it remains to be established whether or not this link is causal. It is speculated that cellular senescence contributes to aging by limiting the numbers of progenitors and stem cells, thus restricting tissue renewal capacity, and that senescent cells secrete factors such as proteases that interfere with tissue function.[150-152] As discussed previously, expression of progerin is associated with heterochromatin disruption and with HGPS aging phenotypes. Progerin expression also results in Notch signaling activity, which causes dysfunction of adult stem cells.[67] These findings suggest that maintenance of heterochromatin may promote longevity by sustaining the renewal capacity of stem cells.

Conclusions and Perspectives

Increasing evidence suggests that the "loss of heterochromatin" model of aging proposed decades ago could perhaps be redefined as the "global heterochromatin loss and redistribution" model, in which a total decline in constitutive heterochromatin occurs together with an increase in heterochromatin at specific loci, contributing to genomic instability and transcriptional changes. The epigenetic alterations resulting from global heterochromatin loss may be at the root of the various molecular events associated with aging and may tie together the various models of aging: the free radical theory, genetically programmed senescence, telomere shortening, genomic instability, nutritional intake and growth signaling (**Fig. 2**).

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